Quiz 2 formula sheet

ECET 250E · 6 November 2026

RL Switch opens or closes · Examples 7.4 and 7.12

1. Find the starting and final currents

Before switching at DC steady state, replace L with a wire. Find the current through that wire. This is I_0, the starting current.

After switching, replace L with a wire again. Find the new steady current. This is If, the final current.

$$I_0 = i_L(0^-) = i_L(0^+), \qquad I_f = i_L(\infty)$$

If the source disconnects and the inductor discharges, If = 0. If it stays connected, solve the remaining DC circuit.

2. Find τ, then write the current

$$\tau = L/R_{\text{seen}}, \qquad i_L(t) = I_f + (I_0 - I_f)e^{-t/\tau}$$

Find Rseen looking into L after switching. Short voltage sources and open current sources.

$$R_{\text{series}} = R_1 + R_2, \quad R_{\text{parallel}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$v_L = L\,\frac{di_L}{dt}, \quad v_R = Ri, \quad v_L = V_s - Ri \text{ (series DC)}$$

RLC First find the two starting values

Current and capacitor voltage

Before switching at DC steady state, replace L with a wire and C with an open gap. Find I_0 through L and V_0 across C. Both carry across the switch.

Example 8.4: no current flows through the 3 Ω resistor before opening, so the capacitor has the same voltage as 6 Ω.

$$I_0 = \frac{10}{4+6} = 1\text{ A}, \qquad V_0 = 6I_0 = 6\text{ V}$$

Choose series or parallel after switching

$$\alpha_{\text{series}} = \frac{R}{2L}, \quad \alpha_{\text{parallel}} = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

α > ω_0 overdamped · α = ω_0 critical · α < ω_0 underdamped.

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

8.4 Series RLC · find current i(t)

1. Find how fast the current starts changing

For the direction shown in 8.4, the capacitor supplies voltage and the resistors use voltage.

$$v_L = V_0 - RI_0 = 6 - (6+3)(1) = -3\text{ V}$$

$$\text{starting current slope} = \frac{v_L}{L} = \frac{-3}{0.5} = -6\text{ A/s}$$

Use the capacitor polarity and current arrow to choose the KVL signs for another diagram.

2. Use the underdamped current formula

$$i(t) = e^{-\alpha t}[B_1\cos(\omega_d t) + B_2\sin(\omega_d t)]$$

$$B_1 = I_0, \qquad B_2 = \frac{\text{starting current slope} + \alpha I_0}{\omega_d}$$

8.4: $\alpha = 9$, $\omega_d = \sqrt{19}$, $B_1 = 1$, $B_2 = \frac{3}{\sqrt{19}}$

8.5 Parallel RLC · find capacitor voltage v(t)

1. Find how fast the voltage starts changing

Use V_0 positive at the top and I_0 downward through L, as shown in her parallel diagram.

$$\text{starting voltage slope} = -\frac{V_0/R + I_0}{C}$$

8.5: $V_0 = 5,\ I_0 = 0,\ C = 0.01 \Rightarrow \text{slope} = -500/R$

Overdamped

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$A_1 + A_2 = V_0$$

$$s_1 A_1 + s_2 A_2 = \text{starting voltage slope}$$

2. Match the damping case

Critically damped

$$v(t) = (A_1 + A_2 t)e^{-\alpha t}, \quad A_1 = V_0$$

$$A_2 = \text{starting voltage slope} + \alpha V_0$$

Underdamped

$$v(t) = e^{-\alpha t}[A_1\cos(\omega_d t) + A_2\sin(\omega_d t)]$$

$$A_1 = V_0, \quad A_2 = \frac{\text{starting voltage slope} + \alpha V_0}{\omega_d}$$

AC Voltage and current · Example 9.8 and RL circuits

Convert the source and solve

$$v(t) = V_m\cos(\omega t + \phi) \quad \longrightarrow \quad \mathbf{V} = V_m\angle\phi$$

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C}$$

Inductor only: $\mathbf{I} = \frac{\mathbf{V}}{j\omega L}$

Series RL: $\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L}$

Add series impedances. Divide magnitudes and subtract angles. Sine → cosine: subtract 90°.

Return to current or voltage in time

$$\mathbf{I} = I_m\angle\theta \quad \longrightarrow \quad i(t) = I_m\cos(\omega t + \theta)$$

$$\mathbf{V}_R = R\mathbf{I}_R, \qquad \mathbf{V}_L = j\omega L\mathbf{I}_L$$

Parallel RL: $\mathbf{I}_s = \mathbf{V}/R + \mathbf{V}/(j\omega L)$

9.8: $\mathbf{I} = \frac{12\angle 45^\circ}{6\angle 90^\circ} = 2\angle(-45^\circ)$ A

For R + jX, magnitude = √(R² + X²), angle = atan2(X,R). Add phasors in rectangular form.